

PROJECT DOCUMENTATION

Student No:	Group No:	Course:
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Phua Sin Yin (s4061241)		
Riordan Cormick-Cox (s4174062)		

Title:

Assignment 2: “Which gaming communities exhibit the most negative sentiment, and what roles do toxic users play within and between those communities?”

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PROJECT PLAN

Project Plan

Start Date: 7 May 2026

Due Date: 25 May 2026

Data Source: [MADOC](#) (Reddit data)

Team Roles & Responsibilities

Member	Main Task
1. Dani Adonai	Sentiment & Topic Analysis
2. Phua Sin Yin	Community Detection
3. Riordan Cormick-Cox	Centrality Measure

Weekly Project Plan

Week	Task Description	Outcome
W1 (7 -17 May)	- Explore Data Source Options - Explore research topic - Finalize initial exploration with tutor - Set up initial code	- Shortlist viable datasets - Tutor approved project topic - Initial code analysis started
W2 (18 – 24 May)	- Align on filtered dataset to use - Map out analysis flow (community -> sentiment -> centrality) - Each team members select main analysis task and begin analysis - Test integration of processed dataset - Prepare final report	- Shared filtered dataset available via Google Drive - Completed each main task assigned - Draft out fully integrated analysis report - Final submission ready

PROJECT PLAN

TIMESHEETS

Member 1: Dani Adonai (s4090448)

Date	Hrs	Work Completed
7/05	1	Joined project GitHub repository, connected with team via Teams, and reviewed assignment requirements focusing on sentiment and topic analysis role.
8/05	2.5	Investigated multiple datasets (YouTube, Kaggle, reddit, x (twitter) MADOC) and evaluated their suitability for sentiment and topic analysis tasks. Coordinated meeting availability with team.
10 /05	1	Participated in team meeting via Teams to confirm project topic and agree on dataset selection for analysis.
11 /05	1	Went over the assignment more and asked question using email to tutor
13 / 05	1	Joined the tutor consultation session online via Teams to clarify project direction and confirm suitability of the sentiment and topic analysis approach
13/ 05	2	Received MADOC dataset via Teams and reviewed its structure for sentiment and topic analysis.
14 /05	1	Planned workflow for sentiment and topic analysis tasks and identified required preprocessing steps prior to team alignment meeting.
15/05	2.5 to 3	Processed ~5,000 dataset entries for sentiment and topic analysis. Cleaned and preprocessed text for modelling Stored outputs externally due to system limits Handled crashes and adjusted pipeline due to large dataset size
17/05	2	Continued refining preprocessing workflow for sentiment and topic analysis after earlier system instability and dataset handling issues.
18/05	0.5	Attended team coordination meeting via Teams to present progress on sentiment and topic analysis and align on integration with other team components.
19/05	2.5	Completed sentiment analysis, topic modelling, and preprocessing work, but needed help from a team member due to a mismatch in requirements and workflow expectations after uploading to GitHub.
20/05 to 22/05	2.5	Continued using sentiment and topic analysis outputs after pipeline was corrected by a team member, and verified preprocessing steps and results aligned with project requirements
22/05	3 to 4	Created the report template in Word and structured the project report and project plan format. Continued working on report writing after confirming details with the team via message.
24-25/05	3	Wrote report sections covering sentiment analysis, topic modelling, and data preprocessing, completed the self-reflection worksheet, attended final team meeting via Teams, and submitted the assignment

Member 1 Self Reflection

Name: ___Dani Adonai_____ ID: __s4090448_____

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Contribution: I worked on sentiment and topic analysis for the project, at first performed data cleaning and preprocessing on dataset of (-5000) which was wrong at beginning. Implemented sentiment scoring and topic modelling (LDA). I tested different modelling approaches and refined the outputs. Created the report structure & template for the main report file and this (project plan)

Challenges: Due the size of the MADOC file my laptop could not handle it, so this caused days of work to fix it first. There was initial confusion with correct pipeline structure which led to some rework (this took time). Also, difficulty aligning sentiment and topic outputs consistently. Confusion about expected output format for report vs code result and unclear integration between my analysis and other team member's work. Work had to be adjusted after feedback from team members to align the structure properly (which good but it took time)

Solutions: Sought help from a team member to clarify and correct the pipeline structure (which I am thankful). I broke the dataset processing into smaller batches to reduce crashes and improve stability. Optimized preprocessing steps to the text processing more efficient. Stored large dataset on my SSD to avoid system limits (which I learned in the hard way). Re-ran section of code step by step instead of full pipeline execution. Used iterative testing, to fix any inconsistencies in my results. Followed team guideline in chat and meeting to ensure my work matches with the assignment needed and overall project structure. Documented each step to not make any additional mistake that could cost me time.

Teamwork Experience: Communicated regularly with the team through Teams' messages and meetings. Received regularly helpful feedback to solve any errors that I have gotten. Shared progress update to ensure transparency of my work (when I was not busy). But at times there were misunderstanding about workflow structure between member (but hey that happens on Uni group work and that is expected). Needed clarification from team member to align my overall project design. Integration between all the different components took too much time and required adjustment after feedback. Some days there was some communication gaps (but I can't blame anyone as all of us have different timetable)

Improvements:

- The group would benefit from agreeing on a unified preprocessing approach early to ensure sentiment and topic modelling outputs remain consistent and easier to integrate.
For large datasets like MADOC, using scalable techniques such as batching or sampling would help improve stability and reduce processing failures.
For future projects, the group should improve early planning and time management by clearly defining the full data processing and analysis workflow before starting implementation.

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TIMESHEETS

Member 2: Phua Sin Yin (s4061241)

Date	Hrs	Work Completed
7/05	1	- Reached out to teammates via Teams - Setup project GitHub page - Read through project specifications
9/05	2	- Research on data source alternatives: YouTube, Kaggle, World Bank Open Data, MADOC - Research on potential topics for research - Schedule meeting time with teammates
10/05	1	- Team meeting to finalise topic and data source
12/05	1	- Attend topic suitability check with consultation tutor
13/05	2	- Download and start on initial data exploration on MADOC dataset with Jupyter notebook - Drafting project research question from initial data exploration
14/05	3	- Understand potential work split structures for project specifications prior to team meeting
	0.5	- Team meeting to decide on work split
16/05	6	- Setup preprocessing of full r/gaming dataset data - Filter for 2020 data rows - Set up community detection code - Refine preprocessing logic before network graph construction - Upload processed data files to google drive (too big for GitHub)
17/05	2	- Run visualization for network community graph - Write community detection report section
18/05	0.5	- Team meeting to show community detection code completed
22/05	6	- Merge community detection dataset findings with Member 1's initial sentiment and topic analysis - Set up 2-way pipeline for data cleaning for topic and sentiment analysis (lesser processing for VADER) - Tested with various LDA params to see if topic analysis findings would be improved - Suggestion for word count frequency topic analysis as LDA does not provide much difference in topic understanding between different discussion communities
24/05	3	- Write report Introduction and Data Collection methods - Write report Approach that summarizes the flow and goal of each member's main task
25/05	6	- Write report network analysis for network construction and overview - Consolidate report conclusion on research question - Write worksheet file self-reflection - Submission and consolidation of code, data files

PROJECT PLAN

Member 2 Self Reflection

Contribution/Reflection: My contribution to this research was mostly in initial data processing, community detection, sentiment and topic analysis, and report introduction and conclusion. With community detection, I had initially assumed that each community would have completely different topics since they were classified as separate communities, but I found that not to be true as discussion networks do not mean separate discussion topics.

Challenges: With the bigger size of our original dataset, analysis code had to be run with incremental saving of analysis progress at various points of code if not the notebook would require a significant amount of time to run through each code change.

Teamwork Experience: Throughout the project, our team has maintained transparent and consistent communication through Teams, and this kept everyone aligned with the project direction and goal. The team initially divided our work based on each analysis method (Community, Centrality and Sentiment), but now in hindsight, we made the mistake of not fully visualizing how the entire data flow would work and how each preprocessing data step would affect another member's work downstream. This led to compatibility issues when moving from community detection to sentiment analysis which required more work on my side to fix for the full sentiment analysis.

Improvements: For future projects, I would consider the data processing flow beforehand to analyze how it could be better shared across tasks and not cause extra hurdles to fix. I would also request my teammates push up their latest changes more frequently, so we can catch coding directions that have gone wrong early instead of discovering issues too late — which in this project required me to redo a significant portion of work. On a positive note, I understand the flow of how our current analysis work together to provide a clear picture of social media interaction relationships and answer our research question.

PROJECT PLAN

TIMESHEETS

Commented [D1]: i think the date for meeting teams , is not right

Member 3: Riordan Cormick-Cox (s4174062)

Date	Hrs	Work Completed
7/05	1	- Connected with teammates via Teams - Setup project github page - Read through project specifications
10/05	2.5	- Researched possible data sources, e.g. Arctic Shift - Meet with team members to consolidate and finalise data source and possible topics - Decide on topic and suggest possible research questions
12/05	1	- Experiment with parquet format and investigate dataset
14/05	1	- Meeting to understand work split - Initial work with centrality section and data requirements/experimentation
15/05	3	- Loading and filtering MADOC parquet dataset for r/gaming to 2020 - Saved filtered parquet
16/05	3.5	- Built directed weighted reply network - Computed degree and PageRank
17/05	3.5	- Computed betweenness centrality - Compiled centrality dataframe - Merged centrality with toxicity/sentiment
18/05	3	- Centrality visualisations - Results exported to CSV and GrapML - Team meeting to go through progress and show centrality code results
19/05	2.5	- Integrate community labels - Community toxicity analysis - Community visualisations
20/05	2	- Community centrality analysis
23/05	7	- Toxicity/sentiment correlation analysis - Write network/centrality report sections
24/05	1	- Work on cleanup for report
25/05	3.5	- Rework community analysis numbers - Report finishing touches - Timesheet/reflection - Meet with team for all the above

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Member 3 Self Reflection

Name: Riordan Cormick-Cox ID: s4174062

Contribution: My contribution to this assignment was network analysis and centrality measures, encompassing both the code and report. This involved loading and filtering our dataset, constructing a weighted directed reply network for the analysis, computing centrality, and toxicity analysis on the userbase and communities.

Challenges: The main challenges I faced were the original exploration of our dataset, which in its unfiltered size cause VSCode to lag and have memory issues, which resulted in the full filtering of the dataset taking almost half an hour, following which the filtered dataset could be used to avoid this issue. Another challenge was the construction of the NetworkX graph, where my initial implementation using pandas iterrows was far too slow to work with, so needed to be reimplemented with itertuples which resulted in a much faster construction.

Teamwork Experience: Transparent and communicative team members made this assignment experience much smoother than it could have been. Regular contact was maintained to ask questions and provide progress reports through Microsoft Teams. There were some challenges with the work split resulting in very individual progress on each section without acknowledging how each of us may need to access the others' work for our own, which thankfully we were able to work through thanks to positive communication streams.

Improvements: For future projects, I would work more closely with my teammates, and have them do the same, to avoid issues with code causing the need for reworking when issues are discovered later than they could be. Furthermore, I would be clearer with the results of my findings and how they might intersect with other sections.